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Unit 4 - Variables

PDF Documents

PDF Document 1: 1738153978105-Chapter 10 - Unit 4 - Variables.pdf

This PDF document is included in the unit materials.

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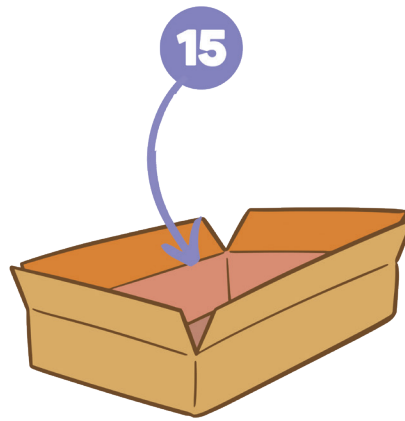
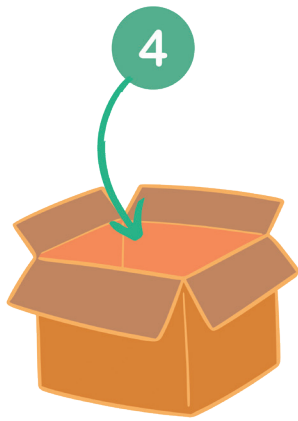


UNIT 4 VARIABLES

Variables

Variables are like labeled boxes in a computer where we keep information we need, like numbers or names. They help us remember and change things in our code, making it smarter.

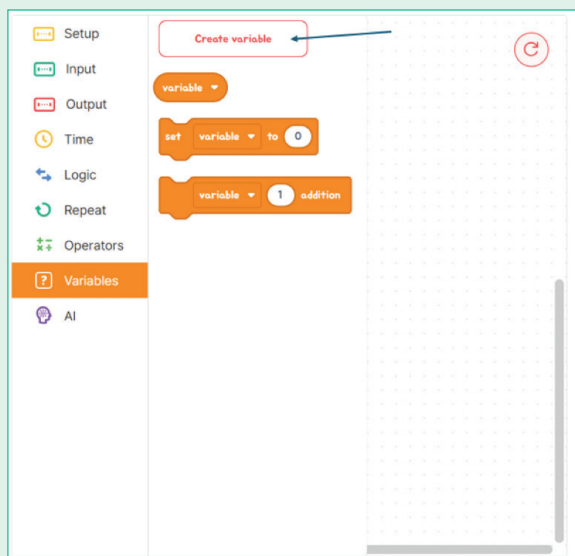
✓ VARIABLES

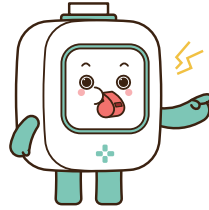


LET'S MAKE A VARIABLE IN CODE EDITOR:

Step1

Go to the Variables section and click "Create Variable."





Step3

After typing the name, click "Make" to create it.

Step2

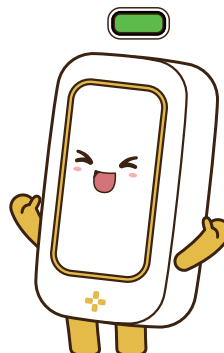
Type the name for your variable (like "mode").

Step4

You'll then see your new variable in the list.

Step5

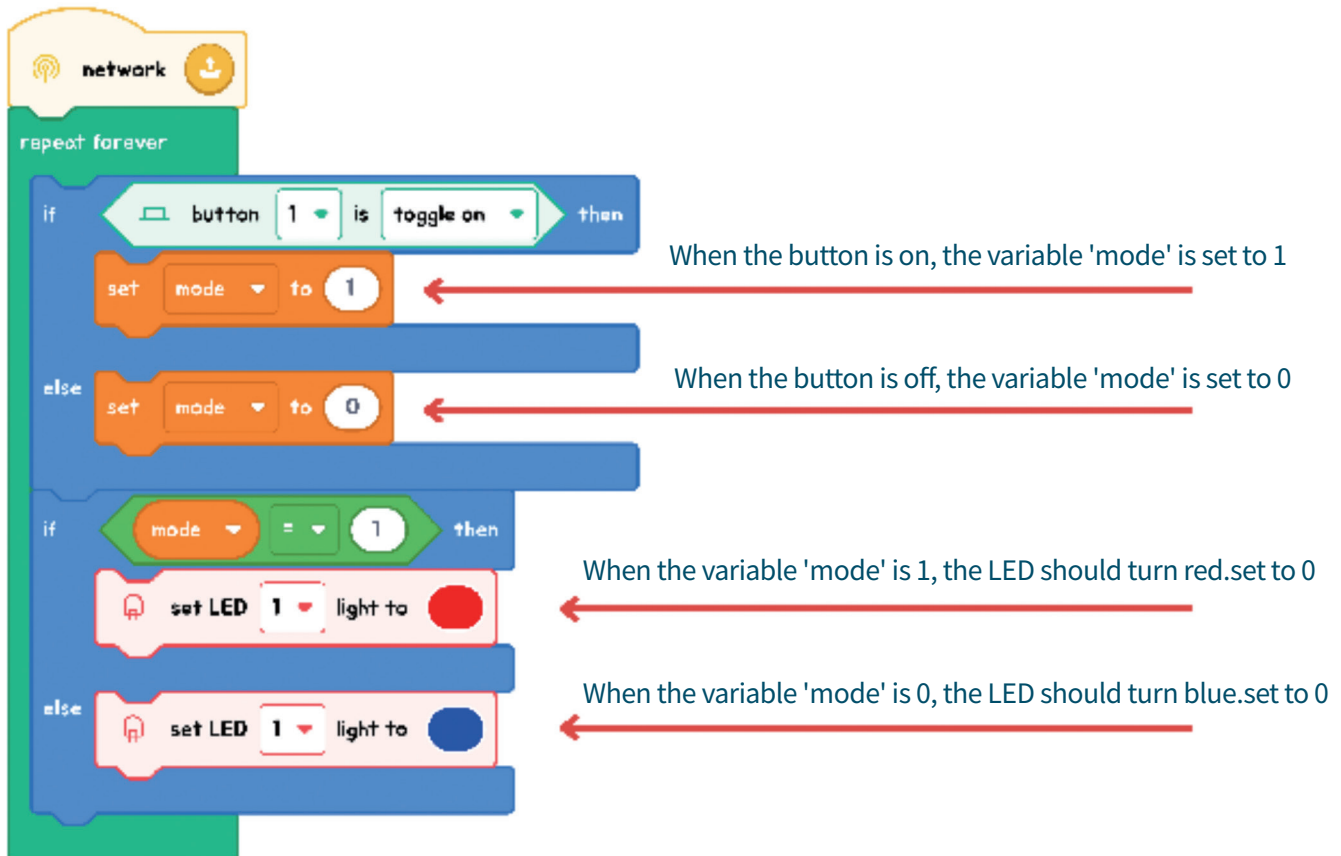
Click on the area where variables are shown to see the one you made.





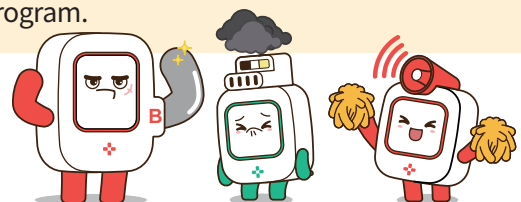
EXPLANATION WITH CODE EXAMPLE

In our code, we're using a button and an LED light along with a variable named 'mode'. Here's what happens in our program:



Code Insight

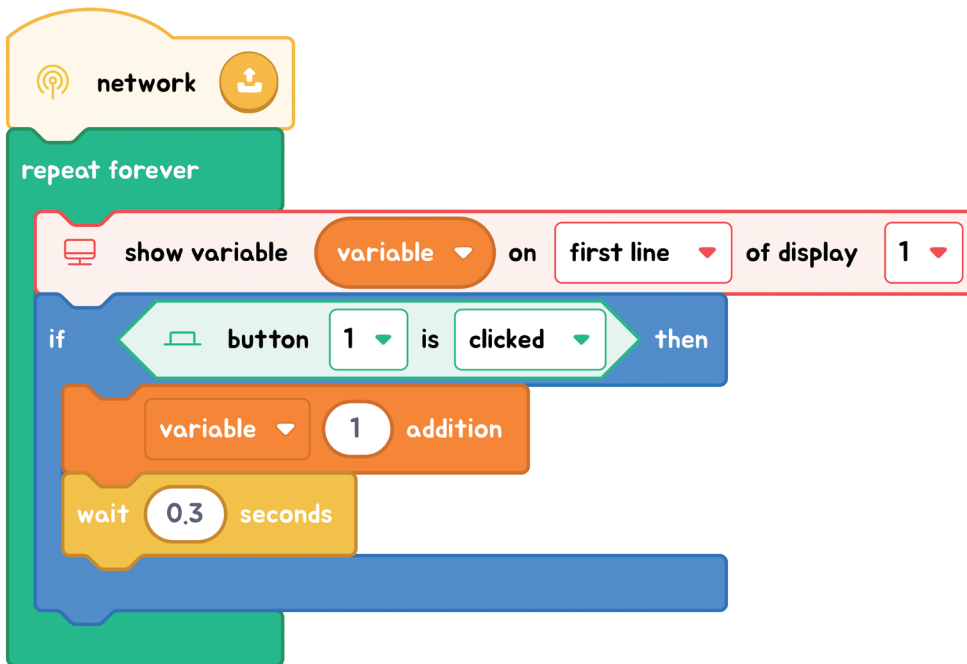
- 1. Variable 'mode':** Think of 'mode' as the program's memory of whether the button is on or off. 'Mode' can be either 1 (on) or 0 (off).
- 2. Checking the Button:** The first part of our code checks the button's state. If the button is on, 'mode' gets set to 1; if it's off, 'mode' goes to 0. This is how the program 'remembers' the button's state.
- 3. Action Based on 'mode':**
 - If 'mode' is 1: The program sees 'mode' is 1 and knows the button was on, so it turns the LED red.
 - If 'mode' is 0: The program finds 'mode' is 0, it makes the LED blue.
- 4. The Role of 'mode':** The variable 'mode' acts as a switch for the LED's color, changing based on the button's state, showing how variables can control actions in a program.



Exercise

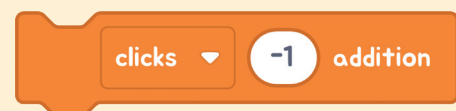
Exercise: Click Counter challenge

Every time you click the button, we want to add one more to our click counter (that's our variable). Using a 'plus' block in block coding, we can keep track of how many times the button has been clicked. When we first make a variable, it starts at 0, unless we tell it to start at a different number.



Code Insight

- Starting Point:** When we begin, **our display screen is set to show a number(variable) called "clicks."** Think of "clicks" like your score in a video game, which everyone can see on the screen.
- Click to Add:** **Each time the button is pressed, we add 1 to our "clicks."** Because the display is always showing "clicks," everyone can see the score go up each time the button is clicked.
- Updating the Score:** Just like how a scoreboard updates every time a player scores in a game, **our display updates the "clicks" number every time the button is clicked, showing the new, higher score.**
- To decrease the number with each button click:** switch to using a subtraction operation in the code.



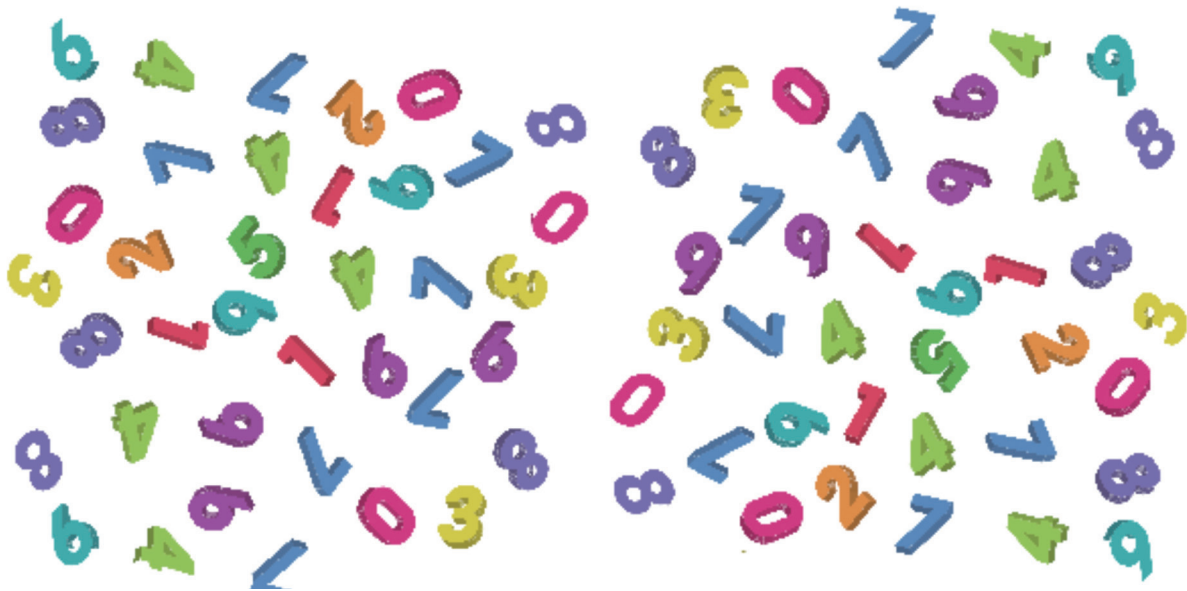


Random

WHAT IS 'RANDOM'?

In programming, when we talk about something being "random," we mean it's like flipping a coin, rolling a dice, or drawing a card from a shuffled deck. You don't know what you're going to get each time. It's the computer's way of making a surprise choice.

✓ RANDOM



HOW DOES RANDOM WORK?

Computers are very good at following instructions, but they don't actually make random choices the way humans can. Instead, they use special commands that create "pseudo-random" numbers. These are sequences that seem random enough for most uses even though they're actually coming from a set pattern.

WHY USE RANDOM?

1. Games

Random numbers can control things like what kind of monster pops out or what treasure you find in a video game, making the game fun and different each time.



2. Simulations

When scientists want to see how things might happen differently, like predicting weather or how traffic moves, they use random numbers to represent all the unknowns.

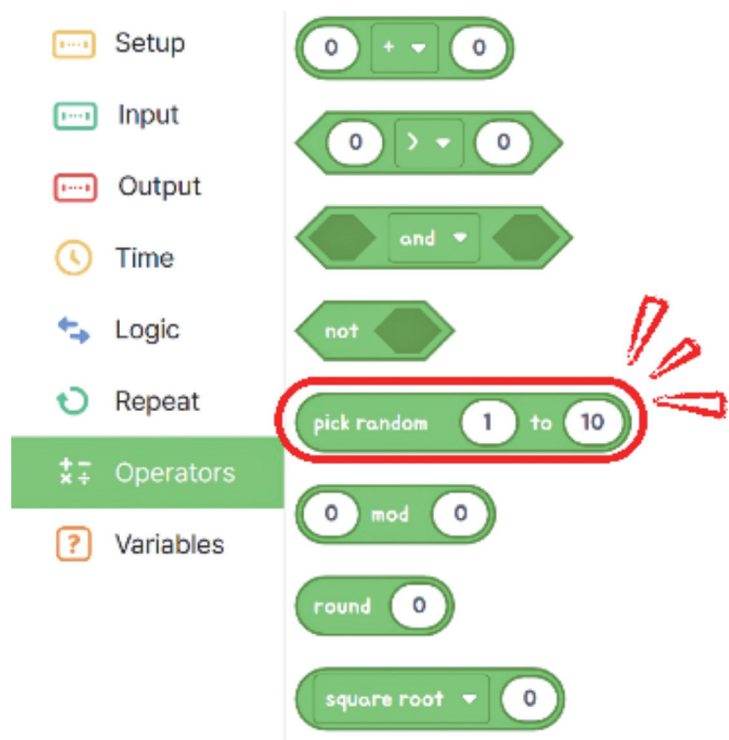


3. Safety and Security

When creating passwords or encryption keys to keep data safe, using random numbers helps make sure they're really hard for someone else to guess.

RANDOM BLOCK IN CODE EDITOR

In the operators section, there is a block for generating random numbers that allows you to specify the desired range.





Exercise

Multiple-Choice Questions



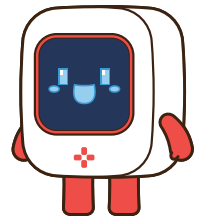
QUESTION 1

When comparing "and" with nested if statements, what similarity do they share?

- ☐ A Both require each condition within them to be checked sequentially.
- ☐ B Both require all conditions to be true for the code block to execute.
- ☐ C Both can be replaced by other logical operators in all scenarios.
- ☐ D Both are primarily used to compare numeric values.

THINK ABOUT IT!

Do you remember?
The nested-if statement
works the same as "AND"!



QUESTION 2

How does the "or" operator differ from the "and" operator in programming logic?

- ☐ A The "or" operator requires only one condition to be true for the code block to execute, while the "and" operator requires all conditions to be true.
- ☐ B The "or" operator is used primarily for exclusivity checks, while the "and" is for inclusivity.
- ☐ C The "or" operator evaluates all conditions at once, while the "and" evaluates them sequentially.
- ☐ D The "or" operator can be used interchangeably with the "and" operator without changing the logic of the code.

THINK ABOUT IT!

What is the main difference
between the "AND" and
"OR"?

